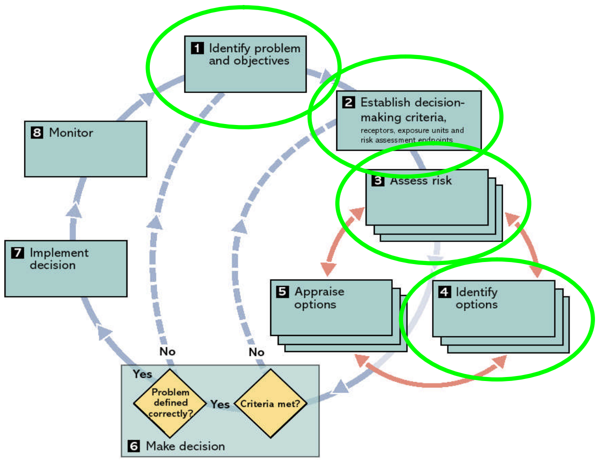
acclimatise
managing your climate risks

The Isle of Man Climate Change Scoping Study

Technical Paper 11

Workshop reports



Report for
Martin Hall, DLGE, Isle of Man Government

Our reference IOM001

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Contents

Introduction

This technical report combines the reports prepared following the two stakeholder workshops held in the Isle of Man in December 2005 and March 2006. The documents record the verbatim comments made by the workshop attendees.

The results from a questionnaire exploring views on developing a formal 'climate change stakeholder partnership' are also included. The questionnaire was distributed to participants at the March 2006 workshop.

Workshop 1 December 2005

Final workshop report

Context

As part of an ongoing climate change project commissioned by the Department of Local Government and the Environment, acclimatise organised a workshop to bring together key stakeholders in the Isle of Man. The workshop was held on Thursday the 15th December 2005 at the Manx Museum, Douglas.

The objectives of the workshop were to:

- Raise awareness of climate change
- Develop an understanding of potential climate impacts
- Identify and agree next steps for addressing climate change
- Begin the process of developing a climate change partnership for the Isle of Man

The agenda for the workshop is included together with a list of participants.

Workshop format

The focus of the workshop was to bring together various stakeholders representing a cross section of key organisations and interests for the Isle of Man.

The day began with a series of short presentations to set the scene by describing some of the research that is currently being undertaken by acclimatise. The remainder of the day was spent discussing the questions posed and sharing information and ideas.

Two workshop sessions were organised during the day, with the participants divided into four groups. A facilitator was provided for each group.

The groups considered eight sectors – chosen to represent the interests of the participants. The sectors considered were:

- Business
- Transport
- Water
- Natural resources
- Infrastructure
- Built environment
- Tourism
- Waste

In the first part of each workshop the groups were asked to look at their sector and identify the key issues, policies and decisions that the Island faces. The participants were asked to focus at the strategic level.

Facilitators were provided with sector overviews to prompt discussion. These overviews were drafted using information assembled for the sector characterisations, which will appear in the final project report.

In the second part of the workshop the participants were asked to identify the key climate risks for their chosen sectors. They were also asked to identify as a group the three most significant risks for the IoM.

A full transcript of the information recorded on flipcharts during this workshop is included.

Significant risks identified by the participants

The most significant risks for each of the sectors were:

- Business
    - Disruption to energy supplies and communications from storm events
    - Impact of sea level rise and storm surge on the location of business property
    - Impact on the perception of quality of life on the IoM
- Transport
    - Impact of winter storms on sea transport, disruption of services, damage to infrastructure and vessels
    - Failure of coastal defences, flooding of roads and disruption to public transport
    - Impact of winter storms on infrastructure, disruption to public transport
- Water
    - Increase in demand for potable water in warmer summers
    - Impact of increase winter rainfall and storms on drainage systems
    - Storm overflows on foul sewerage systems operating more frequently, foul flooding of property
- Natural resources
    - Failure to take a long-term perspective in legislation
    - Lack of baseline data restricts ability to measure progress
    - Limited expertise on the Island
- Infrastructure
    - Impact of winter storms on energy and communications, supply disruption
    - Impact on peak energy demands and changes in seasonal demand profiles
    - Effect on ground conditions and damage to services
- Built environment
    - The group did not reach a conclusion
- Tourism
    - Impact on natural environment
    - Disruption of essential services
    - Impact of storms on travel
- Waste
    - Waste collection and management in higher temperatures, control of vermin and odour control

- Impact of changing rainfall patterns on contaminated land – increase risk of pollution
- Impact of sea level rise on oil storage facilities and contaminated land

Agenda

Department of Local Government and Environment

December 15th Thursday
Manx Museum

Climate change impacts on the Isle of Man
Agenda

Objectives

- **Raise awareness of climate change**
- **Understanding of potential climate impacts in your organisation / area of interest**
- **Identify and agree some next steps for addressing climate change.**
- **Develop an Island partnership on climate change.**

Schedule

Time	Activity	Speaker
0945	*Drinks available*	
1000	**Introduction and welcome**	Martin Hall
1005	**United Kingdom Climate Impact Programme (UKCIP)**	Jacqui Harman
1015	**The Isle of Man Climate Change Scoping Project**	John Firth
1035	**The Climate Science**	Rowan Fealy
1055	**Questions**	ALL
1100	*BREAK*	
1115	**Risk Uncertainty and Decision Making**	John Firth
1135	**Introduction to Workshops**	John Firth
1145 1235	**Workshop session 1** *Sector Issues and Impacts* **Report Back**	ALL
1245	*LUNCH*	
1330 1420	**Workshop session 2** - *Sector Issues and Impacts* **Report Back**	ALL
1430	**Costing of recent extreme events**	Tim Taylor
1500	*BREAK*	
1515	**What next?** **Actions for Progress** **Questions**	John Firth
1530	**Summary and closing remarks**	Martin Hall / John Firth
1545	*FINISH*	

Workshop participants

Organisation	First Name	Surname
Zero Waste Mann	Muriel	Garland
Department of Trade and Industry	Chris	Corlett
Department of Tourism and Leisure	Jackie	Corkhill
Manx Radio	Anthony	Pugh
Ramsey Town Commissioners	Peter	Whiteway
MLC	Claire	Christian
Dept of Transport, Harbours	Ken	Horsley
Isle of Man Meteorological Office	Alan	Hiscott
Isle of Man airport	Phil	Pain
Peel Town Commissioners	Peter	Leadley
Heritage Homes / Dandara Holdings	Ciaran	Downey
Treasury	Mr	Simmim
Manx Telecom	Dave	Comish
Fire Service	Brian	Draper
Manx Bird Atlas	Chris	Sharpe
Manx electricity	Karen	Lesley
Friends of the Earth	Cary	Costain
Agenda 21	Alice	Quayle
Manx Yachting Association	Ron	Spencer
Department of Transport	Peter	Winstantley
Royal Institution of Naval Architects	Peter	Mylchreest
Institute of Marine Engineering, Science and Technology	Jack	Brown
Port Erin Marine Laboratory	Andy	Brand
Isle of Man Post	Malcolm	Macpherson
Agenda 21, Society for the Preservation of the Manx Countryside and Environment, Friends of the Earth	Charles	Flynn
DoEHLE	Jorg	Vanselow
DoLGE	Phil	Styles
IoM Trades Council	Elizabeth	Cueley
Estates Services	Mike	Harrington
Department of Transport	Malcolm	Cowin
Manx Cancer Help Association	David	Gawne
Office of Fair Trading	Ken	Kinrade
Manx Cancer Help	Stephen	Cropper
Manx Gas	Gary	Cregeen
Manx Gas	Nigel	Cross
Manx Bat Group	D J	Blackwell
Manx Bat Group	Mrs G A F	Blackwell
Lloyds TSB	Tony	Wild
Department of Trade and Industry	Martin	Caley
Financial Management	Sally	Roberts

Dept of Trade and Industry	David	Morter
Dept of Trade and Industry	Vicki	Webster
Isle of Man Chamber of Commerce	Barbara	O'Hanlon
Water Authority	Richard	Young
Dept of Agriculture, Fisheries and Forestry	Richard	Selman
Manx National Farmers Union	Ian S	Sleight
Friends of the Glen	Gordon	Elston
Friends of the Glen	Brian	Ballard
	Richard	Hartnoll
DoLGE	Martin	Hall
UKCIP	Jacqui	Harman
Acclimatise	Amy	Hutchins
Acclimatise	John	Firth
University Of Ireland	Rowan	Fealy
Metroeconomica	Tim	Taylor

Workshop transcript

Defining the issue	
Topic	**Built environment**
Question	**Response**
What are the important decisions for the Isle of Man both now and in the future? Is climate already an issue? Is it being taken into account now? Please give examples Do you believe climate change will become a factor or more of a factor?	• **Storm water management of roads** – new developments require storm water management – need to introduce new design standards • **Harbours, sufficient breakwater, quayside height, storm surge height** – take climate change into account for new structures – difficult to do this for existing structures and very expensive • **Traffic congestion, road pollution, need to upgrade train to metro but still able to take trains** – has been considered in the past • **Quality of housing, dampness, housing conditions, flood proofing, alternative energy** – Climate already damp, may get worse in winter. Same or similar climate to Scotland. Implications for building design, Building regulations follow what is happening in the UK, but delayed by 2 years. Dampness is in a problem in existing and old stock, not new stock. Modifications people are making also exacerbates problem. • **Protection of heritage (built and non-built) especially in low lying areas** – 2002 storm surge highlighted current vulnerability. We have not considered the future risk. • **Weirs and reservoirs to control flow and retain water to better manage supply. Need bigger storage for same consumption.** – Quantity in storage is not the problem, getting it to where it is required is. Management and distribution is a problem. Problem is being addressed but not sure if climate change has been included. • **Combined schemes with utilities and government departments, working closer together.** – Is improving – trying to make it happen. Works best on major schemes rather than small schemes. • **Overhead wires going underground.** – Already happening. The driver was damage from storm. Access issues are being resolved. • **Coastal protection of built environment.** – Now includes climate change but cannot protect entire coast line. • **Use Sulby Reservoir for hydropower, with wind power to generate electricity for pumping.** • **Access to minerals at coastal sites, rising sea level.** – Climate change not identified as an issue, no current impact • **Recycling** – Need more recycling of building materials • **Use of renewable energy to reduce reliance on non-renewables** • **Energy efficiency of new buildings** – Driver is to insulate for winter, but get to hot in summer so use energy to cool them. Ventilation is more of a problem. • **Energy inefficiency of old stock** • **Change of building type on farms (e.g. more hot houses) as farming changes** • **More outdoor swimming pools** • **Location of new development**

Climate change risks			
Topic	**Built environment**		
Climate change	Potential risks		
	Design	**Maintenance of existing property**	**Land use planning**
Increasing summer temperatures	Road coating products- tarmac melts Increase cost of new build and upgrade of existing Better insulation	Increased maintenance of road surfaces	Better long term Island planning on how many? Living and working locations and services to support same
Increasing winter temperatures	Increase drain sizes for developments and roads Less need for insulation		
More extreme high temperatures	Increase in building cooling costs Design build to include comfort cooling		
Less extreme low temperatures			
Higher winter rainfall	Water course direction/upgraded Road drains need to be larger	Increased need for maintenance of drainage ditches Risk of flooding to property Change to design criteria for road beds	Covered shopping areas No building in flood plains
Less summer rainfall			Farming outputs decrease Water use management
More intense downpours	Drains and contamination health risk Increase drain size for roads due to increased rainfall	Risk of flooding Extension of separation of storm/foul drainage into separate systems to cope with increased flows	
Sea level rise and increased coastal flood risk		Increased risk of flooding Low lying and harbour side properties at risk	Avoid low lying coastal locations Political influence Harbour and quays
Possibly more winter storms	Need for more storm resistant designs Wave attenuation features offshore from harbours	Keep Manton Road Douglas open instead of closing when snow Risk of damage to existing structures Larger harbour walls and armouring	

Defining the issue	
Topic	**Water**
Question	**Response**
What are the important decisions for the Isle of Man both now and in the future?	Results not recorded
Is climate already an issue? Is it being taken into account now? Please give examples	
Do you believe climate change will become a factor or more of a factor?	
Who are the key stakeholders?	

Climate change risks				
Topic	**Water**			
Climate change	Potential risks			
	Water Supply	**Sewerage and Urban Drainage**	**Land Drainage**	**Coastal Defence**
Increasing summer temperatures	Increase in demand Resource availability	Anaerobic sewage on beaches for longer periods	Reduced groundwater Ground shrinkage Increased erosion on soft cliffs	
Increasing winter temperatures	Shorter recharge season Fewer frozen pipes	Improved sewage treatment	Less risk of snow	
More extreme high temperatures	Increase in demand			
Less extreme low temperatures				
Higher winter rainfall	Quicker recharge Reservoir storage Oil tank structures	Sewer overflows operate more frequently Washout of treatment works Surcharging of sewers	Flooding	Erosion Cliff collapse Increased water table
Less summer rainfall	Reduced yield Demand increases Competition with wildlife for resource	Impact on wildlife Less flooding Less dilution Increasing salinity	Reduction in ground water shrinkage Increase in erosion	
More intense downpours	Polluted run-off	Sewer overflows Lack of storage	Flooding More erosion	More erosion
Sea level rise and increased coastal flood risk		Retaining more water Surcharging Sewer flooding	Flooding Potential pollution from sewage 'Dirty' water irrigation	Increase in wave energy Potential for greater damage Maintenance of flood defences More destructive wave action
Possibly more winter storms		Sewer flooding	Coastal erosion River flooding	Storm damage Erosion

Defining the issue	
Topic	**Waste**
Question	**Response**
What are the important decisions for the Isle of Man both now and in the future?	• **Culture regarding waste management** • **Lack of cost benefit analysis in decisions** • **Economies of scale** • **Changing needs, attitudes and seasonal variation in waste production** • **Export of waste** • **Diversification of the IoM economy** • **Costs to business of waste and waste management** • **Incineration capacity v future recycling** • **Landfills, impact on landscape, risk of pollution form existing and old sites** • **Cost of compliance with new legislation** • **Need for education on recycling v incineration** • **Growth in government sections** • **Waste minimisation, packaging, 'scoop and save' initiatives, need for legislation** • **Shelf life of food** • **Green waste** • **Use of recycled materials** • **Population growth** • **Tourism impacts in peak season creating waste management capacity problems** • **Agricultural practices, impact on baseline environment** **Comment from the group:** Climate change will radically alter our environment. Key will be to ensure local expertise/data exists to target niches within the ever-changing picture. Adaptability.
Is climate already an issue? Is it being taken into account now? Please give examples	• **A little** • **People need to be more aware of the issue**
Do you believe climate change will become a factor or more of a factor?	• **Higher temperatures will create a problem for collection rates** • **Separation of waste**
Who are the key stakeholders?	Business, inhabitants, consumers, DLGE, waste companies, Steam Packet company, MEA,

Climate change risks			
Topic	**Waste**		
Climate change	Potential risks		
	Waste management	**Contaminated land**	**Recycling**
Increasing summer temperatures	Smell Frequency of collection Increasing costs Health Vermin		Need for washing water Decay of green waste Increased volume of cans and bottles
Increasing winter temperatures	Vermin Increase in green waste – extended growing season		
More extreme high temperatures	Vermin Smell Collection disruption Increase costs		
Less extreme low temperatures	Collection disruption		
Higher winter rainfall		Costs of remediation Leaching increases Restoration and use of landfill Future design	
Less summer rainfall		Increasing risk of contamination of drinking water	
More intense downpours			
Sea level rise and increased coastal flood risk		Risk of contamination from oil spillage	
Possibly more winter storms	Backwash of sewers Litter and paper blowing around		Disruption of collection

Defining the issue	
Topic	**Transport**
Question	**Response**
What are the important decisions for the Isle of Man both now and in the future?	Results not recorded
Is climate already an issue? Is it being taken into account now? Please give examples	
Do you believe climate change will become a factor or more of a factor?	
Who are the key stakeholders?	

Climate change risks				
Topic	**Transport**			
Climate change	Potential risks			
	Air	**Public transport**	**Sea**	**Private transport**
Increasing summer temperatures	Load factors on aeroplanes Runway length Increased loads Increasing risk of fog	Comfort Greater use of rail Track failure Walking	Increased leisure activity – safety concerns for boat owners and other users e.g. swimmers Increase in 'water' tourists Water sports Infrastructure concerns – expansion?	Increased use Increase in motorcycle use Road safety
Increasing winter temperatures	Tourism Reduce need for de-icing	No change	Less risk of fog More water based travel	Safer – reduced frost risk Less winter highway maintenance
More extreme high temperatures	Load factors	Engineering issues with increased rail activity 'cold' road clearance	As for previous two sections Access	
Less extreme low temperatures				
Higher winter rainfall	Little effect	Increase due to decrease in walking Flooding		
Less summer rainfall				Flooding
More intense downpours	Flooding	Flooding	Access	Flooding
Sea level rise and increased coastal flood risk		Flooding Failure of coastal defences Service disruption	Stability of structures Links span	
Possibly more winter storms	Service disruption	Damage to infrastructure	Service disruption Damage to infrastructure and vessels	Service disruption

Defining the issue	
Topic	**Natural resources**
Question	**Response**
What are the important decisions for the Isle of Man both now and in the future? Is climate already an issue? Is it being taken into account now? Please give examples Do you believe climate change will become a factor or more of a factor?	• **Protection of countryside from development** • **Impacts on biodiversity –** Already part of modified strategic plan and EIA (but maybe not enough). Zoning is an issue as it is done on a 10year cycle. • **Cumulative effect of loss of small areas of biodiversity and** habitats – Impact being considered in the river environment. Fragmentation from pressure on environment. We already have limited biodiversity • **Negative economic aspects (agriculture and fisheries)** - Agricultural abstraction a marginal issues now, but possibly very big in the future • **Positive economic aspects (agriculture and tourism)** – May loose cold water fish and gain warm, already happening. • **Nature reserve design** – Particularly coastal and high altitude, wetlands etc.. No action on design decisions yet, but already starting to see impacts. Coastal erosion a major issue, need more long term data sets • **How does the Manx community develop – limited space, decisions needed on populations size, where and how to develop etc..** – linked to potential negative impacts with more people waiting to come over, impacts on housing design etc.. Already seeing this due to social change • **Fire management (wildfires, managed burn, higher risk with climate change)** – No climate change impacts yet. Major fire recently – risk will increase. Don't have external resources to draw on like mainland UK • **Energy use for cooling in summer and heating in winter – implications for natural resources arising from use of biofuels and building windfarms** • **Windfarms have an ecological cost in bird strike. Also planning issues e.g. Victorian groups possibly against due to impact on heritage** • **Everything needs to be sustainable and forward looking** • **How we react to climate change will have a big impact on the environment – need to remember this when making decisions** – We are seeing examples of events that may become more frequent • **Need to support local farmers and producers** • **How will trees species change – how will people react to this – we need to plan for the longer term** • **Marine environment – effect of increasing sea temperature on sea life and fishing – the fishing community will be affected** – Algal blooms already affect scallop fishing. We are already seeing changes. Getting warmer water species. Need to maintain research base – should not let this go to Liverpool The workshop group identified a number of issues under a **Governance** heading: • **Need to get a long term perspective in legislation** • **Need to transfer sustainability into action. This is starting to happen in DGLE, but needs to happen in other Government departments** • **Joined up government required** • **Need to measure and benchmark to identify progress. We need more baseline data so we can measure, benchmark progress – there is a problem in getting the suitable people to do this** • **We can adopt methodologies in use in the UK but this is not always appropriate – the IoM is different** • **We need to increase our rain gauge monitoring** • **How much weight do we give to the natural environment when making decisions, particularly when we don't have all the relevant information** • **Tax incentives (and other forms of incentives) to do the right thing** • **Balance between decisions made on financial grounds vs whole costs (e.g. environment, sustainability etc – these require more weight)**

Climate change risks			
Topic	**Natural resources**		
Climate change	Potential risks		
	Biodiversity	**Landscape**	**Agriculture, fisheries and forestry**
Increasing summer temperatures	Change of species indeterminate	More fires because of higher temperatures and less rainfall	Lack of water for crops (grass, fodder, or cash) therefore affecting farming throughout the Island
Increasing winter temperatures	Less winter die-off of sensitive species Bats more active in winter		Disease control on crops more difficult Affects livestock e.g. pneumonia – increasing need for ventilation in farm buildings
More extreme high temperatures	Mortality of animals/plants at extreme of geographical range	Increased risk of heather fires in upland areas Changing tree types in Glens	Increased risk of mass mortalities of commercial species at extreme of geographical range Different crops e.g. vineyards Change in crops and time of harvest
Less extreme low temperatures	Insect pests are a negative but widely observed consequence of biodiversity change Survival of migratory species that currently arrive, breed and die-off e.g. butterflies		Pest die-off reduced in winter
Higher winter rainfall		Soil erosion Flooding affecting erosion on hill land sending terminal moraine into rivers Land drainage/erosion/flooding	Increase in pollutant/nutrient run-off to sea Soil loss from un-vegetated land
Less summer rainfall	Drying out of wetlands in summer Lower summer river flows providing less dilution for existing discharges – river quality problems		Drought affecting farming practices, types of crop grown and livestock reared Adverse affect on marine life
More intense downpours	More frequent sewer overflows causing river quality problems		Damage to plant species More flooding of farmland
Sea level rise and increased coastal flood risk	Increase in salt marsh environment (rare in the IoM) Tidal limits to extend inland, preventing natural flow and dispersion of existing discharges	Coastal defences to be increased 0.5m of sea level rise will jeopardise the IRIS installation on Douglas Prom and all the new apartment dwellings Increased coastal erosion affecting coastal roads Erosion of coastline - loss of land	
Possibly more winter storms		Increased damage/destruction to/of harbour walls Sea water flooding of existing urban areas	More reliance on local food products and strategic supplies

Defining the issue	
Topic	**Tourism**
Question	**Response**
What are the important decisions for the Isle of Man both now and in the future?	• Competition with the landscape and natural resources • Peak tourist season – TT • Visitors v tourists • Repeat bookings • More effort required on product development • Cost of travel to and from the IoM • Demand for large hotels • Need for a diverse tourism industry • Climate not a prime driver for tourism • Marketing • Strategy to attract high income spenders • Standards of service • Training, government sponsored and industry investment • Heritage
Is climate already an issue? Is it being taken into account now? Please give examples	• Yes, but broader issues will affect demand • Water quality impacts • Diversity of wildlife, geology • Debate over marina developments and water based leisure
Do you believe climate change will become a factor or more of a factor?	• Yes, mild weather will increase demand • Need to consider the refurbishment of accommodation • Water supply constraints
Who are the key stakeholders?	

Climate change risks			
Topic	**Tourism**		
Climate change	Potential risks		
	Accommodation	**Nature and marine tourism**	**Related services (eg restaurants)**
Increasing summer temperatures	Air conditioning Extended season opportunity	More marine activity opportunities Water based activities increase Stressful for mammals Drying of moorland – change of vegetation	More transport needed
Increasing winter temperatures		Affect on fisheries	
More extreme high temperatures		Changes in our landscape may affect eco-tourism	Different menus
Less extreme low temperatures			
Higher winter rainfall		Winter holidays more difficult Eden type domes	
Less summer rainfall			
More intense downpours			
Sea level rise and increased coastal flood risk		Impact on marinas	
Possibly more winter storms		Impact on cruising boats and activity holiday	Service disruption Sea travel difficult – boats need to be adapted

Defining the issue	
Topic	**Infrastructure**
Question	**Response**
What are the important decisions for the Isle of Man both now and in the future?	• **Establishing baseline for growth and demand** • **Security of supply – both international and regional** • **Economic supply – energy costs for the producer and the consumer** • **Open markets** • **Competition for telecommunications** • **Increasing regulation/deregulation** • **Consumer perception – energy on demand and cheap**
Is climate already an issue? Is it being taken into account now? Please give examples	• **Fluctuation in temperature has impacts for demand, efficiency, system capacity and financial implications** • **Logistics of securing raw materials and supplies – impact of bad weather** • **Security of supply problems with bad weather** • **Seasonal variation in demand for energy** • **Demand for communications (tele and broadcast) increases in extreme weather** • **Impact of weather on broadcast signal strength**
Do you believe climate change will become a factor or more of a factor?	• **Impact on berthing, loading and storage facilities for oil and gas** • **Government policy (IoM and international) on taxation and regulation**
Who are the key stakeholders?	

Climate change risks			
Topic	**Infrastructure**		
Climate change	Potential risks		
	Gas	**Electricity**	**Communications (telecom, postal, radio)**
Increasing summer temperatures	Out site workers and emergency services – issues pertaining to health and safety Decrease in demand – increase in costs Reduction in sales	Alfresco lifestyle affecting demand Change in demand and change in plant maintenance cycles Increased demand for electricity through air-conditioning	Protection for outdoor workforce More summer visitors – more business Increase taxation to pay for adaptation Increase in summer sickness
Increasing winter temperatures	Security of supply Increase in fuel costs Reduced plant efficiency Underground movement of services	Energy cost fluctuations Increase in demand Theoretical decrease in cost	
More extreme high temperatures	Less demand Equipment failure	Increase demand	Additional air-conditioning increasing need for network services
Less extreme low temperatures			
Higher winter rainfall		Transmission infrastructure failure Xfrs flooding power stations Plant flooding Insufficient pipe capacity	Peak demand during events
Less summer rainfall	Damage to pipelines	Interruptions to water supply and increasing cost of supply Damage to underground cables	
More intense downpours		Transmission and infrastructure failure	Peak demand during events
Sea level rise and increased coastal flood risk	Interruption of gas supplies Flooding of storage compounds	Loss of strategic fuel storage capacity Impact on LPG and fuel oil supplies	
Possibly more winter storms	Supply disruption	Power loss from damage to overhead cables Interruption of oil supplies Loss of supply Continuity of supply	Loss of communications – especially broadcast No standby systems Security of service Damage to overhead cables affecting supplies for communications Increase in demand Reliability of supply

Defining the issue	
Topic	**Business**
Question	**Response**
What are the important decisions for the Isle of Man both now and in the future?	• **Off-shoring – a threat and an opportunity** • **Energy costs to business** • **Maintaining the 'quality of life' for the IoM – gives the Island a competitive advantage – increasing costs, standards for infrastructure and services affecting standard of living** • **Travel costs for personal and business and for goods** • **Delay in communications and physical disruption** • **Relationship with EU** • **Big brother tax position – reciprocal tax arrangements** • **Tertiary level education** • **Retention of skilled people** • **Work permits** • **High level dependence on finance industry – IoM is very vulnerable** • **Lack of competition on IoM between the utilities – high prices**
Is climate already an issue? Is it being taken into account now? Please give examples	• **Severe weather has impacts for financial services and service centres** • **Transport delays – difficulties, disruptions, reliability – problems for perishable goods and delivery of mail** • **Impact on reputation** • **Business continuity planning – some financial service companies are already looking at this** • **Culture regarding waste management**
Do you believe climate change will become a factor or more of a factor?	• **Greater exposure to storms and flooding** • **Impact of climate change on competition, new opportunities for the IoM** • **International regulation and legislation issues** • **Increasing delays for transport** • **Increasing costs and standard of living, making the IoM less attractive** • **Competitive advantage when measured against other off shore financial centres**
Who are the key stakeholders?	

Climate change risks			
Topic	**Business**		
Climate change	Potential risks		
	Economy	**Financial Services**	**IT and manufacturing**
Increasing summer temperatures	Tourism impacts Opportunity to sell the Island – quality of life Increasing risk of fog will have adverse impacts on air and sea travel	Cooling buildings – major problem	Health and safety impacts Environmental health risks
Increasing winter temperatures			
More extreme high temperatures			
Less extreme low temperatures			Impact on workspace
Higher winter rainfall	Adequacy of land use planning policies and building codes Increased risk of flooding in urban areas Disruption to travel – roads impassable		
Less summer rainfall			
More intense downpours			
Sea level rise and increased coastal flood risk	Risks to property – location of development	Major problems for financial services in Douglas Business continuity	
Possibly more winter storms	Property damage Travel disruption Increasing insurance costs		Communications disruption Power outage, disruption of energy supplies

Workshop 2 March 2006

Final workshop report

Context

As part of an ongoing climate change project commissioned by the Department of Local Government and the Environment, acclimatise organised a second stakeholder workshop to bring together key stakeholders in the Isle of Man. This workshop was a continuation of the previous workshop held in December 2005.

The second workshop was held on the 1st March 2006 at the Sefton Hotel, Douglas.

The objectives of the workshop were to:

- Understand potential climate impacts for each stakeholder organisation's area of interest
- Identify impacts and start to look at adaptation options
- Develop an Island partnership on climate change.

The agenda for the workshop is included together with a participant list.

Workshop format

The focus of the workshop was to bring together various stakeholders representing a cross section of key organisations and interests for the Isle of Man.

The day began with a short session to review the impacts from the previous workshop and to identify any gaps. The key issue for each sector was identified and activities that are currently being undertaken on the Island. The outputs from these are within each of the sector summaries below and are reported as verbatim.

Presentations were given on the updated costing work and also on partnership working. A questionnaire was completed; the results from which are can be found on page xx.

The remainder of the day was spent identifying adaptation options for each sector, which are presented in the following sector summaries. The sectors were the same as the previous workshop and continued them through the Risk, Uncertainty and Decision Making process. A full transcript of the information recorded on flipcharts during this workshop is included below in the sector summaries.

In the adaptation options identification workshop, participants were asked to identify adaptation options and to assign the m to one of the following:

- No Regret: deliver benefits>cost whatever the extent of climate change
    - e.g. if already experiencing weather related losses, carry out cost effective actions
- Low Regret: low cost, potentially large benefits under climate change

    - e.g. building climate change in at design stage
- Win-win: provides adaptation AND other benefits
- Flexible / adaptive: incremental, keeping options open

Agenda

Department of Local Government and Environment

March 1st Wednesday, Sefton Hotel. Douglas

Climate change impacts on the Isle of Man

Objectives

- **Review potential climate impacts**
- **Identify adaptation options**
- **Develop an Island partnership on climate change.**

Time	Activity	Speaker
0945	*Drinks available – in the Atrium*	
1000	**Introduction and welcome**	Amy Hutchins / Martin Hall
1015	**Workshop 1: Review of December Workshop and outputs** • Identify any gaps in risk matrixes • Look at priority areas and identify 'hot topics'	Amy Hutchins / ALL
1100	**Update: the Scoping Project and Historic Costing work**	Richenda Connell
1115	**Questions / Stakeholder Questionnaire**	Amy Hutchins
1130	*BREAK*	
1145	**Risk Uncertainty and Decision Making – the Next Steps** • Short presentation to introduce next workshops	Amy Hutchins
1200	**Workshop session 2** • Identifying adaptation options	ALL
1300	*LUNCH*	
1345	**Workshop session 3** • Identifying impacts and adaptation options	ALL
1445	**Stakeholder Presentation** • Presentation on what other areas have done to take other studies forward	Jacqui Harman
1505	***BREAK*** Please complete questions on flipcharts and stakeholder questionnaire	ALL
1520	**Summary and closing remarks**	AH / MH
1535	FINISH	

Workshop Participants

Organisation	First Name	Surname
Met Office	Alan	Hiscott
Design Services Division, Dept of Transport, Director of Design Services	Brian	Cowley
Highways Division, Networks Operations Engineer, Dept of Transport	Bill	Corlett
Harbours, Departmentt of Transport	Capt	Brew
Chief Forestry Officer, Dept of Forestry, Agriculture and Fisheries	Robin	Pollard
Manx Telecom	Dave	Meban
Manx Bat Group	DJ	Blackwell
Manx Bat Group	GAF	Blackwell
Director of Works, Department of Transport	Clive	Callister
Chief Coastguard, IoM Coastguard	Colin	Finney
Western Bee Keepers	Brenda	Stevenson
DHSS	Mike	Harrington
Water Production Supervisor, Manx Water Authority	Martin	Olsen
Water Production Manager, Manx Water Authority	Steve	Knowles
Director of Operations, Manx Water Authority	John	Smith
Networks Manager, Manx Water Authority	John	Leece
Director of Infrastructure Development, Dept of Trade and industry	David	Morter
Environmental Engineer, MEA	Karen	Leslie
Harbours Operations Manager, Dept of Transport	Ken	Horsley
MLC	Clare	Christian
Eaglehurst Ship Management Limited, Technical Director. Royal Institution of Naval Architects	Peter	Mylchreest
Institute of Marine Engineering, Science and Technology	Jack	Brown
Agenda 21, Society for the Preservation of the Manx Countryside and Environment, Friends of the Earth	Charles	Flynn
Dept of Tourism and Leisure	David	Corlett
IOM Post	Malcolm	Macpherson
	Richard	Hartnoll
Planning Manager, Manx Water Authority	Richard	Young
Wildlife & Conservation Officer (Zoologist), Department of Agriculture, Fisheries & Forestry	Richard	Selman
Environmental Protection Officer, DLGE	Phil	Styles
Manx National Farmers Union	Ian	Sleight

Manx National Farmers Union	Ffinlo	Costain
Manx National Farmers Union	J F	Crellin
Port Erin Marine Laboratory, Director	Dr Andy	Brand
Treasury	Sally	Roberts
Manx River Improvement Association	Karen	Galtress
Friends of the Earth	Carrie	Costain
Chamber of Commerce	Barbara	O'Hanlon
DoLGE	Steven	Callister
DoLGE	Martin	Hall
UKCIP	Jacqui	Harman
acclimatise	Richenda	Connell
acclimatise	Amy	Hutchins
Friends of the Earth	George	Uhlenbroek
Friends of the Earth	Philip	Corlett

Sector summaries

Natural resources

Key issue

No co-ordinated environmental regulator. No data collection /monitoring. No budget for doing anything on climate change or learning from others.

Additional Impacts Identified

Higher winter precipitation will mean:

- An impact on seasonal wetlands – perhaps new wet areas. There will be an increase in erosion, mainly in west and northern river valleys.
- Increase in number of landslips on northern hills, which is already being seen as they are on sands and gravels. Need to consider land drainage.
- Higher winter water tables may mean impact on soakaways and septic tanks not working. Already observing higher water tables and impact on septic tanks – as they are not draining away so pools on surface of ground. Note this could be a public health risk. Options could be to improve treatment and pumping systems.

Higher temperatures could mean:

- New pests, an increase in insects leading to an increase in disease risk.
- An increase in insects is good food for bats and fish
- Salmon smoulting(?) earlier and smaller – greater potential for thermal shock when go to sea
- water availability issues for agriculture

Sea level rise could mean:

- Loss of beaches and reduced inter-tidal area, hard defences might prevent their 'migration'
- Salt water intrusion and inundation could affect agricultural land

Changing winter precipitation could mean:

- More intense downpours and droughts between.
- Sulphur oxidised in catchments which is then pushed into rivers and so not seeing benefits of SO2 emissions reductions
- Loose materials put on agricultural lands e.g. muck spreading, fertiliser, seed stock are washed off fields.

Other

- Adaptation in response to climate change could have implications for natural resources .e.g people moving away from flood prone/ coastal areas could end up building in other (natural) areas – who will stop them?
- Enforcement is a major issue / monitoring. There is only one planning enforcement officer.
- Changing vulnerability / desirability of different locations and impacts on natural resources.

Current activities on the Island

- Data is already being collected on SST, salinity, nutrient data. There are some very long records and some go back 100 years – so mustn't stop them now as need them most. Monitoring is especially good when the environment is changing fast.
- DAFF just started bird monitoring
- Change in bee behaviour has been observed
- There is a sustainability working group which is currently developing a 'sustainability toolkit' for all government policies and projects.
- Some places already can't get insurance because of flooding.
- Subsidence is also a problem in some places
- Planning system is working better than it used to
- Housing pressures are mainly for affordable housing. Some pressure on the countryside but not actually that bad. Most of the building is in town and village extensions. Rural housing has affordability issues e.g. for farm workers.

Natural resources adaptation matrix

Adaptation Type	**Urgent: Opportunities to take advantage of now / decisions needed now / cost effective now**	**Long term**	**Keep a 'watching brief'**
Share loss	• If insurance is required, encourage the financial sector to examine if there is a good business opportunity for IoM insurers		
Bear Loss	• Joined up thinking across government and treasury (low regret) • Create an environment agency / climate change department with a budget (low regret) • Manage retreat encourage new habitats and increase biodiversity (Win win) • Scallop fishing (increase in sea temperatures are associated with algal blooms) • Do not sanction additional hard defences in non urban areas – likely to adversely affect natural resources.	• Ecological change / biodiversity (win win)	• Adopt forest management systems to accommodate storm ? (No regret) • Increase in variability of river flow, less diversity inverts less fish and the associated knock on effect on ecosystem (no regret)
Prevent effects: structural / technological	• Fire plans – these are already been done • Agricultural land use adaptation (win win)	• Buildings to be adapted • Improve riverbank management to provide some counteraction to climate change impacts on habitat quality and availability • Agricultural related infrastructure may need to evolve to match changes in land use. (NO regret / win win)	
Prevents effects: Legislation / regulation / institutional	• Eco tax • Daff, DLGE, Island Agenda 21 have role to play in developing All Island Strategic plan (win win) • All Island Strategic plan needs to take account of climate risks e.g. floods and changes to agriculture (win win) • 'Sticks and carrot' need to change behaviours e.g. building regs on loft installation plus enforcement	• Environmental laws • Plan for habitat change in site protection (flexible) • Fire risk preparation – agree access routes for fire engineers, supplying first defences (e.g. fire beaters) (no regret)	• Population increase
Avoid / exploit changes in risk	• Raise profile of issue and implication to land managers to allow evolution of management (win	• Adapt forestry to suit new migrating birds	• Flooding • Agriculture: new crops will

	win) • Exploit opportunities stemming from climate change e.g. grow exotic for a – not oaks • Upgrade existing surface water drainage systems and rivers to deal with significant rainfall events and flash floods • Agriculture land adaptation – have to adapt by Jan 2011 due to introduction of free 'derogation' markets (can't limit imports anymore) and see if subsidy regime can help promote change e.g. stewardship	• New product market development to reflect changes (win win)	need new infrastructure processing plants – and a new market. Can't talk about adapting agriculture without adapting the associated infrastructure. (win win)
Research	• Species and habitat monitoring - set up central biodiversity centre (no regret) • Maintain and monitor terrestrial, fresh water and marine environmental data (no regret). This also forms part of wider network e.g. UK and Ireland. Irish sea co-ordination will be lost with other studies e.g EU projects (as IoM are not eligible for these alone). • Government needs to take the lead and get better involved in other for a e.g. BIC • Initiate (continue?) coordinated monitoring programs of environmental factors under single control with dedicated budget • Effects of increased water table on lowland environments • Bird monitoring (just started) No regret • Need To monitor river flow rates; phytophthora, groundwater levels, species taking over. Eg. Bracken encroaching upwards on moor land. The frost keeps it off the top of the hills at present but this may change. (No regret) • Need all island monitoring system and central collection point, which is computer, based. Co-ordination of agencies and volunteer agencies to agree on central system (no regret) • As a small island with limited resources need to prioritise on monitoring e.g. indicator species for each habitat.	• Research on forest / tree pests / diseases so can adapt tree species choice. (win win)	
Educational /	• Voters need to know what government is investing	• Extend business planning process	• Include in all departments

Behavioural change	in e.g. adaptation to climate change and how they can support it. • Need to educate MHKs on climate change, use Manx radio and explain relevance to the island. Need to lengthen their outlook horizon • Food mile labelling required so people know where food comes from and can buy in an informed way • Need to engage treasury and explain economic implications of climate change and get them to consider environmental and social issues and not just cost. • Undertake cost benefit analysis on major capital projects and take account of climate change . Design climate change in early to save £. Maybe use lifetime costing to account for full costs.	• Promote all projects and expenditure – at point of development(e.g. on a new coastal defence 'built as part of climate change response') • Provide incentives for saving water e.g. reductions on rates for households with water saving devices in w.cs • Food miles labelling • Assess impact of climate change on employment sectors and adapt tertiary education to meet expected skills in long term	business planning process (low regret)

Tourism

Key issue

Island becomes less attractive / beautiful. Loss of amenities e.g. beaches

Additional impacts identified

Higher winter precipitation will mean:

- Hotels on Douglas promenade already have damp basements / cellars and are prone to flooding
- Cultural sites such as Rushen Abbey and Castletown could become more vulnerable as in flood prone areas.
- Don't have so many tourists as footpath erosion is an issue

Higher temperatures could mean:

- Greater demand on water resources from TT races as more tourists.
- Increase burden on health service from tourists – of sunburn and exposure to heat.
- Increased fire risk – already lost two mountains sides – Bradda Mountain; Carnanes – was supposed to be controlled heath land fire, which got out of control. Not good at coping with wildfire.
- Implications for emergency services
- Air quality concerns as increase in asthma

Sea level rise could mean:

- Loss of beach – Port Erin
- Loss of Douglas Promenade
- Loss of heritage sites especially Castle Mona Hotel
- Castle town on the Links course there is a possibility of saline intrusion.

Changing precipitation could mean:

- Decrease in water quality

Increase in storminess could mean:

- transport problems which could put tourists off. Ships and catamarans are now 'low loaders' which are more prone to problems in sea conditions

Other

- Island becomes less attractive / beautiful. Loss of amenity e.g. beaches maybe not always less attractive – just different.
- Basking sharks generate tourism, however will there be there under climate change.
- Tourism – deep sea fishing is popular – how will this change

Current activities

- DTL / MNH market research
- Calf of Man has existing H20 availability problems – there are isolated affected places – not widespread
- Already monitoring toxic algal blooms and shellfish poisoning – could become an issues for tourists eating shellfish.
- The main scallop fisheries are here and can get closed due to algal blooms, which would be very costly.
- Jellyfish – most poisonous species here as cold water but some of the species are warm water.

Comments

- Most options put in the 'Long –Term ' box
- Opportunities to plan ahead, less criticality than for natural resources.
- Tourists don't visit in anticipation of glorious sunshine
- People are coming here for short stays due to 'feel of the place' – an island; natural environment.
- 1/3 visits are to friends and relatives, which should be unaffected. Travel costs are a key issue.
- Golf – usually have own water supplies of ground water and hence they are self-sufficient.
- Water resources shouldn't be a problem at all because they have three years storage. Issue is investment in distribution that is being addressed.
- 'Business tourism' Isle of Man is adaptable, independent and autonomous and could become an exemplar for the rest of British Isles. This is aspirational at the present, but big changes like in agriculture in 2011 could drive innovation.

Tourism adaptation options

Adaptation Type	Urgent: Opportunities to take advantage of now / decisions needed now / cost effective now	Long term	Keep a 'watching brief'
Share loss		• Unreliable ferry service in variable weather (low regret)	
Bear Loss		• Hotels – air conditioning etc. (flexible) • Exploits increased summer temps – produce more visitors – economic benefits (win win) • Loss of amenity some irreversible – specific beach locations – specific building. Maintain and adapt as long as possible.	
Prevent effects: structural / technological		• Develop / expand facilities for water based recreation. Options will improve (win win) • Plan for effective water supplies under drier summer conditions (win win) • Design engineering solutions to address risk of transport links being disrupted by flash floods / erosion • Allow for sea level rise in harbour development (low regret) • Specialised holidays by changing land use and diversifying into habitats for key species. (Win win) • By exemplifying best practise we can build business tourism to see how we're tackling the issue (win win) • Encourage improvements to buildings to cope with excessive heat in summer and / or wet weather in winter	

Prevents effects: Legislation / regulation / institutional	• Protection of landscape From erosion / fire disaster • Encourage crop rotation • IoM airport runway extension – build to allow further extension (low regret) • Develop TT short course to reduce effect of disruption due to hill fog	• Provide incentives for more environmental friendly take away food packaging (low regret)	
Avoid / exploit changes in risk	• Health risks / bad publicity from toxic algal blooms and shellfish (currently monitored) • Agricultural diversification – 'holiday lets' moving into this more.	• Make best use of landscape ecological change to maintain recreational use of the landscape. Adapt locations and activities (win win) • Motor sports may need to refocus for weather infrastructure reasons	
Research	• Monitor changes to identify and promote new opportunities • Landscape assessment (MNK planned ?) (No Regret)	• Golf research (win win) • Warmer summers encourage sea bathing (win win) • Yachting business • Specialised holidays (flexible) • Impacts on ecotourism (win win) • Research of tourism market futures for natural and cultural tourism (no regret)	
Educational / Behavioural change			•

Transport

Key Issue

Severe weather impacts on both sea and air, which are a short term to repair, and land, which take longer time to repair

Additional Impacts Identified

Higher winter precipitation could mean:

- The stability of highways is more susceptible to slips etc

Current activities

- Integrated transport group – not sure if this includes impacts and adaptation
- Dropping large stones on Gansey beach to protect road i.e. maintenance of existing coastal protection applies to many areas.
- All Island Strategic plan addresses transport issues – this covers a range of sectors and includes mitigation and adaptation
- Tidal flooding research is going on to provide better warning systems. Sand bags are provided when flooding is expected.

Transport adaptation options

Adaptation Type	Urgent: Opportunities to take advantage of now / decisions needed now / cost effective now	Long term	Keep a 'watching brief'
Share loss	• Agree roads that can be allowed to be lost or are too vulnerable to be maintained and step protecting in maintaining them • Ensure strategic spares in stock for linkspans and other critical equipment to minimise disruption		
Bear Loss		• Consider moving coastal roads, Manx electric railway further inland to avoid risk of coastal erosion	
Prevent effects: structural / technological	• Improve land drainage to prevent flooding / damage to roads • Has EIA on runway extension taken into account impacts of climate change (no Regret) • Where financially viable enhance river banks to contain flood water (low regret)	• Provide more suitable vessels better adapted to cope with expected increased in winter storms (e.g. bigger seacat) • New road schemes to have improved drainage systems e.g. Peel Road • Upgrading of railways to use for commuters • Provide better harbours facilities with more sheltered berths and approach to take into account more inclement weather events	• Review previous flood events and reinforce road / drainage infrastructure to cope. • Upgrade railway link from Ramsey to Port Erin to modern fast, comfortable and low floor recess metro system. And also include freight capability.
Prevents effects: Legislation / regulation / institutional	• Higher taxation on 'gas guzzling' vehicles. Tax on CO2 output, more tax on multiple cars per household • Implement integrated transport document	• Minimise car use by relocation of some government offices and encourage electronic conferencing • Relocation of business to other towns	• Put freight on to railways. Recreate integrated rail system – a modern version, gives alternative to road
Avoid / exploit changes in risk	• Encourage use of electric cars and more fuel efficient ones • Use TT course as testing ground for solar power and other non-fossil fuel vehicles (win win)	• Aircrafts – instrument landing systems – use technology so planes can land in cloud. • Prepare plans and equipment to tackle storm events more effectively	
Research	• Survey the coast. Identify strategic locations that cold be vulnerable to	• Can farmers grow crops for bio fuels • Research into electric cars and other fuels	• Integrated transport policy – realistic but not idealistic

	damage due to age, conditions etc (no regret) • Cost benefit analysis of houses in flood plains and of measures needed to mitigate damage (no regret) • Improve flood alerts, storm warnings	• Use of railways as commuter transport brings in other options for moving people in the event of loss of roads in storm events • Minibus circular routes to help tourism and holiday makers in scenic areas (as tourism increases)	• Need to improve public transport in general
Educational / Behavioural change	• Better public transport, commuter trains from Port Erin to Douglas. More effective links from main bus routes. • Education for children on future environment (Low regret) • Greater use of public transport / car sharing	• Less reliance on 'One person, one car' syndrome.	

Water and waste

Key issue

Waste Key Risk: Impact on contaminated land – increased risk of pollution

Water key risk: Impact of increase winter rainfall and storms on drainage systems

Additional impacts

Higher winter precipitation will mean:

- Business continuity resilience – keep reservoirs high versus risk of flooding
- Hydro plant run all year – opportunity

Higher temperatures could mean:

- Changes in demand for water
- Effect on water resources as temperatures increase and seasonal runoff pattern changes
- Increase in biological treatment – opportunity

Changing precipitation could mean:

- Changes in demand for water
- Storm water / combined sewer system and surcharging – has impact on flooding, water quality, coastal and local watercourses.
- Supply versus compensation flows – impact on ecology and habitats
- Extraction from the south – issue of moving water and possible saline intrusion

Increase in storminess could mean:

- Structural impacts on tall buildings / structures e.g. incinerator

Current activities on the Island

- Flood mapping research
- Iris scheme
- Low flush 'blocks' for toilets
- Drainage areas plans – show capacity of existing sewer systems to take the load
- Surface water separation schemes are ongoing on existing systems. They are mandatory in new development to have separate systems.
- Water meters – for commercial. There are few in domestic properties
- Provision in new builds to be able to put in meters in boundary boxes so can be installed at a later date.
- Sulby flood alleviation scheme

Waste and water adaptation matrix

Adaptation Type	Urgent: Opportunities to take advantage of now / decisions needed now / cost effective now	Long term	Keep a 'watching brief'
Share loss	• Surface water separation schemes (win win) • Increasing capacity of the sewerage systems (no regret) • Increased refuse collections – to reduce health risk as temperature rises • Maintenance of existing drainage systems (Low regret) • Greater demand on recreational use of higher ground / reservoir. Increased costs of mitigating pollution risks	• More reservoirs for storage (no regret)	
Bear Loss	• Identify all houses in flood plains – buy and demolish • Water butts for garden use (win win)	• Pulrose power station - risk of flooding. 1953 flood removed bridge	• Old landfill sites – pollution , leachate, erosion,.
Prevent effects: structural / technological	• Flood defences, structural recommendations on all new developments to reflect flood alleviation risks	• Increase in capacity storage	• On landslips
Prevents effects: Legislation / regulation / institutional	• One waste authority on the island. • Increased segregation of waste at source. • Set up environment agency to regulate • Compensation water regulations • Planning control on development in flood plains and areas susceptible to coastal areas	• Catchment management for island's rivers (low regret) • Sustainable urban drainage systems (low regret) • Promote agri – environment schemes (low regret)	• Legal requirements for water meters. • Controlled retreat of coastline
Avoid / exploit changes in risk		• Increase use of sewage in agricultural irrigation	
Research	• Flood defence lessons from uk and other similar issues / islands • Improve monitoring of rainfall and runoff	• Identify potential flood plains – areas of risk and assets at risk	• Risk of geosmin and algae pollution of impounding reservoir. Additional treatment costs involved.
Educational /	• Shelf life is decreasing and supply chain		• Changing waste profile e.g. more

Behavioural change	demands are increasing. • Encourage support of local produce (no regret) • Metering to reduce water consumption • Water conservation by user education (win win) • Education needed on waste separation and segregation (no regret)		green waste and increased rate of frequency of collection

Business and infrastructure

Key issue

Impact on winter storms on energy and communications – supply disruption
Effect on ground conditions and damage to services

Additional matrix impacts

General impacts that:

- Business continuity / disaster recovery is an opportunity to exploit – need regulations / legislation in place
- Transport needs business continuity for goods in and out, individual needs and an increasing population demand.
- Opportunity that tourism will increase as southern Spain becomes less attractive.
- Getting staff / logistics in extreme events (e.g. MEA)
- Increased erosion on coast.

Activities ongoing on the Island

- Suncream provided as part of health and safety
- Town centre in Ramsey is designed at a certain level above sea level.
- Adding storm water events as part of development e.g. larger pipes. This is an island wide policy.
- Energy audits are being undertaken (ISO14001)
- Reduction in leakages and other losses from system (this has more than halved).
- DLGE is doing development into solar gains
- Flood map research by HR Wallinford – also flow gauges are being installed.

Business and infrastructure adaptation options matrix

Adaptation Type	Urgent: Opportunities to take advantage of now / decisions needed now / cost effective now	Long term	Keep a 'watching brief'
Share loss	• Increase tax on property owners who live in area of risk	• Develop alternative energy sources	• Other islands are at greater risk – watch and learn
Bear Loss	• Business continuity requirements (win win)	• Stockpile essential supplies (e.g. oil)	
Prevent effects: structural / technological	• Risk assess existing assets to identify investment required (win win) • Build houses on stilts • Diversification of business by utilities • (flexible) • Undertake flood risk assessments on all new builds	• All electrical systems should be put at dado rail levels • Put cables underground and not overhead	
Prevents effects: Legislation / regulation / institutional	• Immediate legislation to ensure all business and infrastructure issues have climate change as a key issue • Lifetime costs through planning / building regs e.g. drainage specifications (win Win) • Change building regs so that properties can withstand more extreme conditions	• Building energy conservation (Design and installation) (no Regret) • Legislation – the Climate act • Government strategy on how to deal with loss of footpaths when erosions happens.	• Planning / building design requirements (low regret)
Avoid / exploit changes in risk	• Link river monitoring to all island alarm system (link flood defence and tides to same alarm)	• Sustain island's growth to afford changes e.g., creativity in business, tax and incentives. • Use solar power as reduction in cloud cover • Improve extreme weather reporting – advanced warning system	
Research	• Improve monitoring of rainfall and runoff	• Develop heat exchange systems • Use of tidal variations to generate tidal power • Alternative energy sources	
Educational / Behavioural change	• Issue sun cream to staff • Education of energy conservation • Increase awareness within all sectors of business • Energy audits of business	• Cancer awareness training • Change to working conditions – education • Working conditions – new clothing / sunscreen (share loss)	• Change of working hours (no regret) • Plant trees to provide shade (win win)

acclimatise
managing your climate risks

Built environment issues

Key issues

Management of storm water, sewerage and other services in high rainfall and storm events.

Additional impacts

Higher winter precipitation will mean:

- Flooding of low lying coastal areas following high tides / tidal surges

Increase in storminess could mean:

- Communication infrastructure e.g. masts

Other

- Emergency storage for fuels
- Population expansion will lead to housing encroaching on even more open land than it already does
- Protect island beauty from coastal erosion

Activities ongoing on the Island

- Some separation of storm and other waste required on new buildings
- Building regulations have been updated to include various things
- Planning more friendly on solar and wind energy applications
- River and coastal monitoring / understanding projects.

Built environment adaptation matrix

Adaptation Type	Urgent: Opportunities to take advantage of now / decisions needed now / cost effective now	Long term	Keep a 'watching brief'
Share loss	• Weir and reservoir to better manage water flow and retain more water for better management of supply. • Coastal protection of built environment	• Ensure major items of infrastructure are protected against storm / flooding damage. Where these events are considered few and far between and it would not be cost effective to replace them e.g. linkspan	
Bear Loss			• Stop development in coastal erosion zones • Let nature take it course in coastal zones
Prevent effects: structural / technological	• New harbour / coastal structures to take account of sea level rise and increased storms • Change drainage design standards to allow for future demand (Low regret) • Storm water separation from foul sewerage system	• Improve maintenance of infrastructure to deal with climate change e.g. improve capacity of rivers / drainage systems by cleaning • In Commercial buildings have less retention on heat retention and greater emphasise on design for 'cooling' systems	
Prevents effects: Legislation / regulation / institutional	• Flood proofing of housing. • Do not allow house building in likely flood areas / coastal erosion • Revise plans sooner to prevent building on flood plains • Review Island Strategic Plan to take into account the effect of climate change (No Regret) • Regulation and inspection must take place in building trade to ensure energy efficient methods used • Creation of an 'environment agency' to ensure new legislation is carried out properly with proper inspectors methods and if necessary prosecution • Redirect resources from planning / visual	• Offer grants to improve energy efficiency in existing old houses	

	impact to planning design / building control		
Avoid / exploit changes in risk		• Solar and wind power maybe viable • Changes for agricultural crop rotation and farming methods • More tourists but bigger drain on resources	• Use Sulby reservoirs pumped storage for instant power availability for peak loads and emergency electric supply
Research	• Understand situation of dampness, housing conditions and whether they need 'flood proofing' • Identify those roads that are most prone to flood damage (historical data) and produce action plan (no Regret) • Need more information on rivers and flows • Keep existing places of research open and running to monitor on going changes in the environment (no regret)		
Educational / Behavioural change	• Education and behavioural changes is thought most important by group • Audit business, households and ministry to reduce energy use • Campaigns now to educate people in how their lifestyles contribute to decreasing climate change (low cost) • Educate children, adults and make information on risky areas widely available		

Note: Salt marsh option is not viable on the Island as there has been a PhD on this.

Name one action that you feel really needs to be done, who should it and when

- Everybody –ecological footprint.
- Identify funding for partnership – DLGE
- Increase awareness – DLGE / partnership
- Water management improvements
- Big campaign on educating public / business on reducing their CO_2 emissions i.e. less waste of energy

From the workshop what is the one thing that your organisation should be doing?

- Introducing life time costing into treasury appraisals
- Awareness – assess impacts on plants / process
- Pressuring government to consider all aspects of climate change – especially environmental impacts and mitigation issues
- Agricultural land – use adaptation
- Food miles labelling
- Continue public campaigns to raise awareness of what they can do in their households / lifestyles to reduce CO_2 emissions now.

Summary of responses to the 'partnership' questionnaire

Isle of Man Climate Change Partnership – what do you think?

The first part of the questionnaire asked the workshop participants to indicate their level of interest in an Island climate change partnership. Out of a total of 30 questionnaires received back – the overwhelming majority (21) said they wanted to have an active engagement in a partnership.

21	I am interested in having an active engagement in an Island climate change partnership (i.e. through contributing to its development, providing guidance, expertise and contacts, shaping its work programme, etc.)
7	I am interested in receiving information on the work of an Island climate change partnership and am willing to react to that information
1	I am only interested in receiving information on the work of an Island climate change partnership
0	I am not interested in an Island climate change partnership

Questions

1. Do you want a partnership on the Island?

25 people said 'Yes'
No response '3'; Other '2'.

'Other' responses included:

- Not standalone – it must be linked to another area such as Scotland
- Only if manageable team size – if >6 or 8 then unlikely to focus correctly

Quote: "It is essential!"

2. What would you want an Island Climate Change partnership to achieve? What would be its principle functions and activities?

- Provide focus
- Cross fertilisation of ideas between organisation / joined up thinking between government and industry
- Ensure climate change is considered in plans / strategy
- Education / awareness

- Forum for discussion and public debate
- Influence government / policies
- Undertaking projects
- To manage its own running budget
- Risk analysis
- Building networks
- Share knowledge / source of information

3. What do you want the intended focus to be? Adaptation or adaptation and mitigation?

Adaptation and mitigation : 17
Just adaptation: 13

4. Who would be the members of the partnership and its network? Which key organisations do you consider must be involved if a partnership is to be successful?

- Representatives of business
- Cross government departments
- Interested groups
- Open to all?
- Main services
- Local communities
- All at the workshop
- Climate change experts
- Politicians
- Chamber of commerce
- Relevant NGOs

5. Who will be the lead agencies:

Treasury / DoLGE / DOT

Main consensus was that Treasury should fund it with DoLGE to lead it.

Suggestion: 'Partly funded to start the process but an independently funded body eventually'

6. How would you like the partnership to be run? e.g. steering group / meeting of all stakeholders / informal contact by email? Other suggestions?

Majority indicated that the steering group should lead and drive forward with full stakeholder meetings / workshops as necessary. A number of comments covering the benefits of group working "'it is a big benefit in all getting together'

7. Do you anticipate a role for a full time co-ordinator / secretariat?

   Majority (16) said 'yes' , 5 responses said that it was part of DoLGE's function. 3 responses indicated 'Not yet' and only 3 said 'No'.

   Funding could be an issue – a suggestion was made that a part time co-ordinatoes may be more appropriate, or alternatively DoLGE should provide a resource.

8. Where would the partnership be resourced / financed / funded from?

   Majority indicated it should be government funded – with a suggestion that the Treasury should provide the funding, on its own or together with other key agencies.

9. Who should manage the development phase between now and the partnership?

   Overwhelming said that DoLGE should lead and drive forward

10. Please provide any other suggestions for the Island climate change partnership

   - Need to set measurable targets and actions
   - Need to look at short term horizons not just 2080 view
   - Key input into planning / utility decisions
   - Take good ideas from elsewhere and adapt them
   - Need input from economic stakeholders
   - Co-ordination essential

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